ATLAS Operator Manual

Installation, Setup, and Operations

Autonomous Telescope & Learning Astronomy System

Version: Design Phase 1.0
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Chapter 1 — Introduction

ATLAS is autonomous observatory control software for amateur astronomers who want their nights to produce real scientific work. It orchestrates a complete observing session — from pre-flight checks through dawn flats and post-session processing — using five specialised AI agents working in a clear chain of command.

The Mission

Most amateur observatories produce beautiful images. ATLAS is built to also produce data that contributes to professional science: precise astrometry of minor planets, photometric measurements of variable stars and exoplanet transits, and the early discovery of supernovae and other transients. Every clear night can produce a real scientific deliverable.

The Five Agents

Agent	Role
Planner	Strategist — builds nightly schedules and NINA sequences
Critic	Watchdog — continuous monitoring, never decides
Operator	Authority — final say on every autonomous decision
Archivist	Historian — post-session processing and reporting
Oracle	Researcher — anomaly and transient detection, knowledge management

NOTE All five agents run on the Anthropic Claude API (claude-sonnet-4-6). No local AI models. An internet connection and an Anthropic API key are required for agent decision-making. When connectivity drops, ATLAS falls back to a deterministic safe-autonomous rule set until the API is reachable again.

Chapter 2 — System Requirements

Observatory PC

- Windows 10 or Windows 11, 64-bit
- Minimum 8 GB RAM (16 GB recommended)
- Storage for the reference frame library and session data. 500 GB minimum; 1–6 TB recommended depending on cadence
- Stable internet connection (drops are tolerated, but long outages reduce autonomy)

Required Third-Party Software

- **NINA** 3.x with the **Advanced API** plugin (Christian Palm)
- **PHD2** guiding software with the JSON-RPC server enabled (port 4400)
- **ASTAP** plate solver with a star catalog installed

Optional Third-Party Software

- **Siril** — deep-sky stacking pipeline (free, open-source)
- **AutoStakkert!4** — planetary lucky-imaging stacker
- **SharpCap** — high-frame-rate planetary video capture

Required Accounts

- **Anthropic API key** — for the five AI agents
- **ntfy.sh topic** — for alert push notifications (free, or self-hosted)

Optional Accounts (for science submissions)

- **AAVSO Observer Code** — for variable star and exoplanet photometry
- **MPC Observatory Code** — for asteroid/comet astrometry submissions
- **TNS API Token** — for supernova/transient candidate submissions

Chapter 3 — Installation

ATLAS ships as a single installer for the observatory PC. The installation is silent — no questions, no decisions — and handles every dependency automatically.

What the Installer Does

- Installs a private Python runtime and all required libraries
- Creates the SQLite database with the full schema and seed data: seasonal target catalogs, agent system prompts, default safety thresholds
- Loads all five agent definitions
- Configures the FastAPI server on port 5000
- Creates two desktop shortcuts on the observatory PC:
- **Start ATLAS Observatory** — launches the backend server and all agents
- **Open ATLAS Dashboard** — opens the web dashboard in your default browser

What the Installer Does NOT Do

- Configure your hardware (telescope, camera, focuser, mount, dew heater)
- Set your site coordinates
- Connect to NINA or PHD2
- Store your API keys or submission credentials

> **NOTE** All hardware and site configuration happens after installation, through the Setup tab in the dashboard. This is intentional: the same installer works on any observatory anywhere in the world.

First Launch

1. Double-click **Start ATLAS Observatory**. A minimal console window appears confirming the server has started on port 5000.

2. Double-click **Open ATLAS Dashboard**. Your browser opens to `http://localhost:5000`.

3. On first launch, the dashboard automatically takes you into the Setup Wizard (Chapter 4).

Chapter 4 — First-Time Setup Wizard

The wizard walks through every configuration ATLAS needs to operate. Each step has a **Skip / Manual Entry** escape hatch if you prefer to fill values directly.

Step 1: Master Password

You create a master password. ATLAS uses it to derive an encryption key that protects all stored credentials (API keys, submission tokens). The password is never sent anywhere and never written to disk in plain text.

Step 2: Anthropic API Key

Paste your Anthropic API key. ATLAS validates it by making a test call to the Claude API.

Step 3: Site Coordinates

Enter your latitude, longitude, and elevation. The wizard can auto-detect from your IP as a starting point, but you should verify against your GPS or surveyed pier position.

Step 4: NINA Connection

Enter your NINA host (typically `localhost` or `127.0.0.1`) and port (default 1888). ATLAS attempts a connection and confirms the Advanced API plugin is responding.

Step 5: PHD2 Connection

Enter your PHD2 host and JSON-RPC port (default 4400). ATLAS attempts a connection and confirms PHD2 is responsive.

Step 6: Equipment Profile

Declare your camera type (One-Shot Colour or Mono+Filter Wheel), filter names if applicable, telescope focal length and aperture, sensor pixel size, and any specifics ATLAS needs for plate solving and photometry. You can also let the wizard auto-derive pixel scale by plate-solving a test image.

Step 7: Mount Capabilities

Indicate whether your mount supports non-sidereal tracking. Asteroid and comet workflows will be disabled if not, and the Planner will avoid scheduling them.

Step 8: Roof Configuration

Choose one of three roof modes:

- **NINA-controlled** — ATLAS commands the roof through a NINA-supported dome driver
- **Custom-driver** — use the documented extension interface to plug in your own roof driver
- **Manual** — ATLAS waits for you to open and close the roof; alerts fired at session start and end

Step 9: Submission Credentials

Optionally enter your AAVSO Observer Code, MPC Observatory Code, and TNS API Token. These are needed only for the corresponding science workflows. Stored encrypted.

Step 10: Notification Setup

Enter your ntfy.sh topic name and severity preferences. ATLAS sends a test push to confirm.

Step 11: Run a Simulated Night

The wizard offers to run a full simulated session immediately, using fake hardware responses and synthetic frames. This verifies every part of the pipeline before your first real night.

> **TIP** You can revisit any wizard step at any time from the Setup tab. Run the simulation as often as you like — there is also a permanent **Run Simulation** button in Setup.

Chapter 5 — The Dashboard

The dashboard runs in any modern browser on the warm-room PC (or any device on the LAN). Six tabs cover what an operator actually does.

Tonight

The live session view. Large GO/NO-GO banner at the top — green, yellow, or red — with the Operator agent's plain-language verdict and reason. Status lights for every subsystem (mount, camera, focuser, guiding, weather, internet, power). Current target, sub progress, guiding RMS graph, live preview. The **Take Control** toggle is in the top bar and visible from every tab.

Plan

Active campaigns, the Planner's schedule for tonight, Oracle's candidate suggestions. Approve, reorder, or reject targets. Add ad-hoc targets.

Science

The submission queue — every candidate measurement awaiting your approval before going to MPC, AAVSO, TNS, or NASA Exoplanet Watch. Each candidate shows the data, the fit, the residuals, and a one-click approve/reject.

History

Past sessions, full HTML reports, target history, lifetime totals, all-time scientific contributions log.

ATLAS

Text and voice chat with the Operator agent. Voice uses the browser Web Speech API — no install, works on any modern browser. Ask questions, request status, give natural-language commands.

Setup

All configuration: wizard, hardware, site, equipment, credentials, retention policies, notification preferences, the **Run Simulation** button.

The Take Control Toggle

A persistent toggle in the dashboard top bar. While engaged:

- All five agents pause command actions (they continue monitoring and logging)
- Direct hardware controls appear on the Tonight tab: slew, capture, focuser, filter wheel
- Session state is held — the autonomous sequence can resume from exactly where it paused
- Every action you take during manual control is logged into the session record for the report

> **WARNING** Releasing manual control returns full authority to the Operator agent. If conditions have changed during manual operation, the Operator may re-evaluate and request a schedule rebuild from the Planner.

Chapter 6 — A Typical Night

Pre-Session (Automated Pre-Flight)

Before the Operator commands the roof to open, it runs a complete pre-flight checklist:

- NINA connected and responsive
- PHD2 connected and responsive
- Camera connected, cooling reaches setpoint
- Focuser connected, current position valid
- Mount connected, parked at known position
- Filter wheel connected (if applicable)
- Recent darks exist for tonight's exposure plan
- Recent flats exist for tonight's filters
- Disk free space > configured threshold
- Weather GO for next 60 minutes
- Internet up (or safe-autonomous mode armed)
- Claude API responsive
- Calibration freshness within the 7-day window
- Power source nominal (not battery with low runtime)

Any failure produces a no-go with the specific reason surfaced on the dashboard and via ntfy.sh push. You can override individual checks if you know what you're doing.

Session Run

Once cleared, the Operator commands the roof open, the Planner pushes the NINA sequence, and the session begins. Critic watches continuously — 90-second fast loop, 5-minute standard loop. Any anomaly is reported to the Operator, which decides whether to continue, pause to standby, or escalate to you.

Monitoring from the Warm Room

The Tonight tab gives you everything at a glance. Walk in, look at the banner. Green: go back to bed. Yellow: read the reason. Red: ATLAS has paused; check the alerts.

Dawn Flats and Close-Out

As dawn approaches, the Operator hands the schedule to its dawn-flats routine, captures sky flats for each used filter, then closes the roof, parks the mount, and warms the camera at the configured ramp rate. Finally, it triggers the Archivist.

Post-Session

The Archivist runs calibration, registration, stacking (Siril for deep-sky, AutoStakkert!4 for planetary), photometry, and astrometry on the captured frames. It generates the full session report and notifies the Oracle of new data. The Oracle runs the transient pipeline against the night's frames and adds any candidates to the Science tab's submission queue.

> **TIP** In the morning, you'll typically check three things: the GO/NO-GO history, the session report on the History tab, and the submission queue on the Science tab.

Chapter 7 — Science Workflows

ATLAS organises science work into **campaigns**. A campaign has a goal, a target or target set, a success criterion, and a cadence. Campaigns persist across nights, weeks, or months. The Planner draws from active campaigns when building each night.

Asteroid & Comet Astrometry

Create a campaign with the target's MPC designation. ATLAS resolves the ephemeris, computes non-sidereal tracking rates, slews, captures, plate-solves, performs the astrometric measurement, and produces an MPC-format report. Success criterion is typically a number of astrometric epochs. Submission queued for your approval.

Variable Star Photometry

Create a campaign with the AAVSO designation. ATLAS pulls the comp star sequence, performs differential photometry, generates AAVSO-format observations. Multi-night campaigns (e.g., monitor Betelgeuse V-band every clear night for one year) are first-class objects.

Exoplanet Transit Photometry

Provide the TIC/HAT/WASP designation and predicted transit windows. ATLAS schedules the relevant nights, observes fixed-field with no dithering, produces a calibrated light curve in NASA Exoplanet Watch or AAVSO Exoplanet Section format.

Supernova / Transient Survey

Define a field list or a galaxy catalog. ATLAS visits each field on a configurable cadence, builds up reference frames (three visits minimum), then runs image subtraction on subsequent visits. Candidates surface to the Science tab with cutouts and catalog cross-match results. Approved candidates are submitted to the Transient Name Server (TNS).

Planetary Imaging

Schedule planetary sessions by target. ATLAS launches SharpCap, configures ROI and high frame rate, captures SER video. AutoStakkert!4 stacks the best frames. Output is a high-resolution planetary image plus session metadata.

Deep-Sky Imaging

Standard deep-sky workflow with full calibration, Siril stacking, and proper photometric and astrometric metadata. Pretty pictures fall out as a by-product of the science pipeline.

Chapter 8 — Submissions

Every scientific submission queues for your approval. ATLAS never auto-submits. Real-name attribution matters and the human stays accountable.

The Submission Queue

Found on the Science tab. Each pending submission shows:

- The candidate identity (target name, position, type)
- The measurement data, fit, and residuals where applicable
- Catalog cross-match results (for transient candidates)
- A cutout image and comparison to reference
- The exact text or structured data that will be submitted
- **Approve**, **Reject**, or **Hold for Review** buttons

Submission Targets

Workflow	Submission Destination
Asteroid & comet astrometry	IAU Minor Planet Center (MPC)
Variable star photometry	AAVSO International Database
Exoplanet transit photometry	NASA Exoplanet Watch / AAVSO Exoplanet Section
Supernova / transient candidate	Transient Name Server (TNS)

WARNING The TNS in particular has strict policies around accidental and duplicate reports. ATLAS will refuse to submit a candidate without a satisfactory catalog cross-match (typically against Gaia DR3, Pan-STARRS, and the MPC). When in doubt, hold for review.

Chapter 9 — Simulation Mode

Simulation mode lets the full ATLAS pipeline run without commanding any hardware. All five agents operate normally; NINA, PHD2, and ASTAP calls are intercepted and return synthetic responses. The Archivist and Oracle run on fake frames.

When to Use It

- After installation, to confirm every part of the pipeline works
- After a configuration change, to verify nothing broke
- Before a major upgrade, as a regression check
- To rehearse a science workflow without burning a clear night

How to Run It

1. Open the Setup tab.

2. Click **Run Simulation**.

3. Choose the simulation profile (single short session, full night, transient detection drill).

4. Watch the Tonight tab. The session runs at accelerated time — a full night completes in a few minutes.

5. Review the simulated session report on the History tab when finished.

> **TIP** Simulation mode is the right answer to “Will ATLAS handle this scenario?”. Run it whenever you have a question.

Chapter 10 — Safety Systems

Calibration Freshness

Critic continuously checks the calibration library. Darks and bias frames older than the freshness window (default 7 days) trigger a warning; significant temperature drift from the dark library setpoint also triggers a warning. Flats are evaluated per-filter.

Weather Monitoring

Critic pulls live weather (Open-Meteo by default) every 5 minutes during operation. Configurable thresholds for wind speed, humidity, dew point margin, and cloud cover. Threshold breach generates an alert to the Operator, which decides between continue, standby (light or full), or emergency shutdown.

Emergency Shutdown Sequence

1. Stop imaging immediately
2. Park the telescope (confirmed via mount feedback)
3. Close the roof (if automated)
4. Warm the camera at the configured ramp rate (default 5 °C/min)
5. Power down hardware (focuser, dew heater, mount motors)
6. Save session state
7. Notify the operator via ntfy.sh push (critical priority)

Power-Loss Handling

ATLAS detects the active power source via the OS UPS interface. Setups with multiple sources — solar/battery, utility, generator — each register as a source. The Operator agent tracks transitions and runtime estimates. Default shutdown trigger: 50% remaining battery *or* 5 minutes runtime remaining, whichever fires first. Tunable in Setup.

> **NOTE** Once shutdown completes, ATLAS does not auto-resume when power returns. The operator must explicitly approve a resume, even if conditions are otherwise favourable.

Chapter 11 — Storage Management

ATLAS tracks disk usage continuously and enforces operator-configured retention.

Defaults

- Raw sub frames: kept 90 days after stacking
- Planetary video files: kept 30 days
- Session reports: kept indefinitely
- Calibration masters (bias, darks, flats): kept indefinitely
- Reference frames (transient library): kept indefinitely — protected from cleanup
- Submitted measurements and approved candidates: kept indefinitely

Alerts

- Warning at 80% disk full
- Refusal to start a new session at 95% disk full
- Cleanup preview before any deletion runs — shows exactly what will go

> **TIP** On smaller installations, configure raw sub retention down to 30 days. The science deliverables (measurements, reports, references) take very little space; raw frames are the bulk consumer.

Chapter 12 — Updates

ATLAS checks daily for new releases on its GitHub releases page. Updates are never installed without your consent and never interrupt a session in progress.

What Is Preserved Across Updates

- Your database (sessions, campaigns, target history, science contributions)
- Your configuration (site, equipment, credentials, retention policy)
- Your reference frame library
- Your calibration masters

Update Process

1. ATLAS notifies you via the dashboard banner when an update is available
2. Click the banner to see release notes and review the changes
3. Click **Install Update** at a time of your choosing (never during a session)
4. The installer runs silently, preserving all configuration and data
5. ATLAS restarts and confirms the new version is operating correctly

> **WARNING** Major version updates (e.g., 1.x → 2.x) may require additional configuration steps. Always read the release notes before installing.

Chapter 13 — Troubleshooting

NINA not responding

Verify NINA is running and the Advanced API plugin is enabled. Confirm the host and port in Setup. From the observatory PC, try opening `http://[NINA-host]:1888/v2/api/equipment/camera/info` in a browser; you should see a JSON response.

PHD2 calibration fails

Check the guide camera connection, look-ahead time, and that PHD2 has a clear sky with guide stars. ATLAS will attempt recalibration twice before escalating.

Claude API unreachable

ATLAS automatically falls back to safe-autonomous mode: continue current target, hold the schedule, no replans, no risk decisions. Full agent control resumes when the API is reachable. Verify your Anthropic API key is valid and you have remaining quota.

Plate solve fails

Confirm ASTAP is installed and the star catalog is present. Verify pixel scale and FOV in your equipment profile match reality. A single-shot plate solve from the Tonight tab will give the exact error message.

Internet down

Imaging, guiding, and plate solving continue normally. Catalog lookups fall back to the local cache. Submissions queue until connectivity returns. Agents fall back to safe-autonomous mode.

UPS triggering shutdown

Review the power thresholds in Setup. Confirm the OS sees your UPS (Windows Power settings). If your power configuration has multiple sources (battery, utility, generator), ensure each is registered with the OS.

Dashboard won't load

Confirm **Start ATLAS Observatory** is running. From the observatory PC, try `http://localhost:5000`. From the warm room, use the observatory PC's LAN IP. Check the Windows firewall is not blocking port 5000.

Chapter 14 — For the Curious: How It Works

Architecture

ATLAS is a single Python process running a FastAPI web server on port 5000. The server handles three things: serving the web dashboard, exposing a REST API for the dashboard's use, and running the five AI agents as concurrent asyncio tasks. All state lives in a SQLite database accessed via SQLAlchemy.

Agent Message Passing

Agents communicate through structured message objects on internal queues. Critic posts alerts to the Operator's queue; Operator posts revision requests to the Planner; Operator posts post-session triggers to the Archivist; Archivist posts new-data notifications to the Oracle; Oracle posts candidate-target proposals back to the Planner. All messages persist in the database for audit.

Database Schema (Overview)

- **sessions** — every observing session, state, conditions, decisions
- **campaigns** — multi-night research efforts with goals and progress
- **targets** — every object ATLAS knows about, with knowledge threads
- **frames** — every captured frame, with full FITS metadata and quality grade
- **measurements** — photometric and astrometric measurements
- **submissions** — the submission queue and history
- **references** — the transient reference frame library
- **credentials** — encrypted API keys and submission tokens
- **alerts** — every alert raised, severity, and outcome
- **messages** — inter-agent message log

The Workflow Pipelines

Each science workflow has its own pipeline of stages: target resolution → sequence build → acquisition → calibration → measurement → submission preparation. Pipelines are declarative; adding a new science workflow means defining its stages, not modifying the agents.

Appendix A — NINA Configuration

Required NINA Plugins

- **Advanced API** by Christian Palm — the API ATLAS uses for all hardware control

Advanced API Settings

- Enable the Advanced API server
- Bind to the LAN IP of the observatory PC (or 0.0.0.0 for any interface)
- Set the port to **1888** (default)
- Confirm the API base path is `/v2/api`

Verifying NINA Is Ready

From any browser on the observatory PC, the URL

```
http://localhost:1888/v2/api/equipment/camera/info
```

should return a JSON response. If it returns 404 or fails to connect, the Advanced API plugin is not enabled or not bound correctly.

> **NOTE** Use the observatory PC's LAN IP rather than `localhost` if you encounter binding issues. EmbedIO (the HTTP server NINA uses) sometimes does not bind to `localhost` properly.

Appendix B — External Service Registration

Anthropic API Key (required)

Visit `console.anthropic.com`. Create an account, generate an API key, and ensure your account has billing enabled. Paste the key into Step 2 of the Setup Wizard.

ntfy.sh Topic (required for alerts)

Visit `ntfy.sh`. Choose a unique topic name (treat it like a password — anyone who knows the name can read your alerts). Install the ntfy mobile app on your phone and subscribe to your topic. Enter the topic name in Step 10 of the Setup Wizard.

AAVSO Observer Code (optional)

Visit `aavso.org`. Register as an observer; AAVSO assigns you a multi-character observer code. Enter it in Step 9 of the Setup Wizard if you intend to do variable star or exoplanet photometry.

MPC Observatory Code (optional)

Visit `minorplanetcenter.net`. Apply for an observatory code by submitting astrometric measurements of known objects from your site. Once assigned, enter the three-character code in Step 9 of the Setup Wizard.

TNS API Token (optional)

Visit `www.wis-tns.org`. Register as a user, create a TNS bot under your account, and generate the API token. Enter it in Step 9 of the Setup Wizard if you intend to submit supernova/transient candidates.

Appendix C — Glossary

AAVSO	American Association of Variable Star Observers.
Astrometry	Precise measurement of the position of celestial objects.
Campaign	A multi-night research effort with a goal, target, success criterion, and cadence.
Critic loop	The continuous monitoring cycle run by the Critic agent (90 s fast, 5 min standard).
Differential photometry	Measuring a target's brightness relative to comparison stars in the same field.
FITS	Flexible Image Transport System — the standard astronomical image format.
FWHM	Full Width at Half Maximum — a measure of seeing/focus quality.
Image subtraction	Subtracting a reference image from a new image to reveal changes (used for transient detection).
MPC	IAU Minor Planet Center — the international clearinghouse for asteroid and comet astrometry.
NINA	Nighttime Imaging 'N' Astronomy — the imaging control software ATLAS uses for hardware control.
NEOCP	Near-Earth Object Confirmation Page — MPC's list of objects needing follow-up.
OSC	One-Shot Colour — a colour camera using a Bayer matrix sensor (vs. mono+filter wheel).
PHD2	Open-source autoguiding software.
Photometry	Measurement of the brightness of celestial objects.
Plate solve	Determining the WCS (sky coordinates) for an image by matching stars to a catalog.
Reference frame	A deep stacked image of a field, used as the comparison for image subtraction.
Safe-autonomous mode	ATLAS's deterministic fallback when the Claude API is unreachable.
Standby	A paused state where ATLAS holds position and waits. Light standby keeps cooling; full standby powers down.

Submission queue	The list of approved-pending candidate measurements awaiting operator approval before external submission.
TNS	Transient Name Server — the international clearinghouse for supernova and transient candidate reports.
WCS	World Coordinate System — metadata mapping image pixels to sky coordinates.